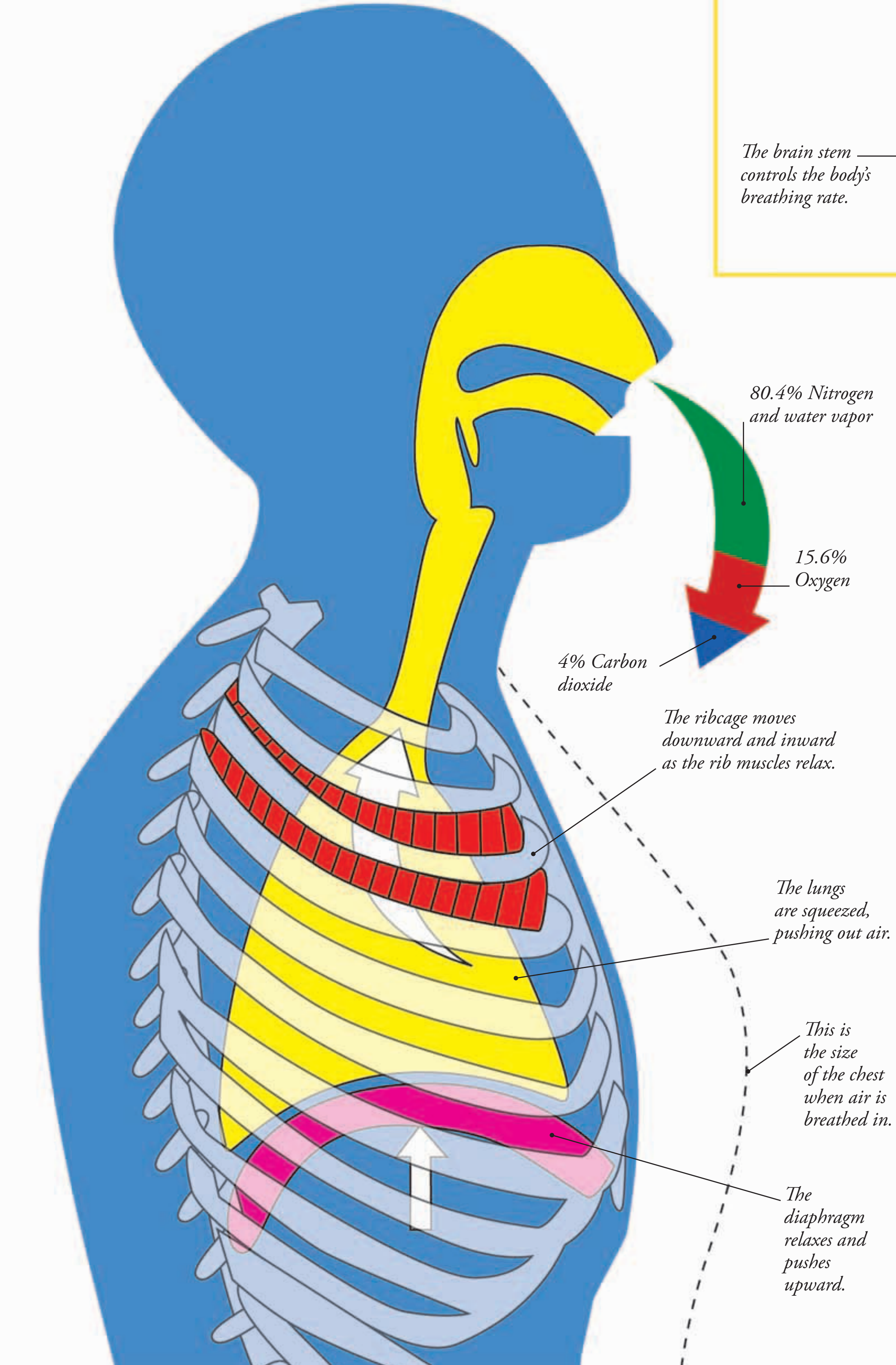
AIR OUT

When you breathe out, your diaphragm relaxes and springs back into its natural, curved shape, pushing up against your lungs. At the same time, your rib muscles relax and let your ribcage drop back downward and inward. As a result, your lungs are squeezed and air is pushed out through your mouth and nose. The air you breathe out contains less oxygen and more carbon dioxide than the air you breathed in.

CONTROL CENTER

You don't have to think about breathing. The brain stem (orange) at the base of your brain (green) automatically controls your breathing rate. This control center monitors the amount of carbon dioxide in your blood. If the carbon dioxide level rises, during exercise for example, your breathing rate goes up to supply more oxygen to your muscles and to flush out the excess carbon dioxide.

The brain stem controls the body's breathing rate.



FAST OR SLOW?

At rest, we breathe in and out between 12 and 15 times a minute. During exercise, such as running, our breathing rate can more than double and we breathe more deeply. This is because our muscles are working harder and need more oxygen to release the energy required for movement. They also produce more carbon dioxide, which must be removed.